

CHARLES K. GIFFORD

Glen Rock, New Jersey 07452 | ckgifford06@gmail.com | (201) 983-8795

[Linkedin](#) | [Github](#) | charliegifford.com

EDUCATION

Boston College, Chestnut Hill, MA

Expected: May 2028

Bachelor of Science: Computer Science; Minor: Mathematics

GPA: 3.33

Honors: Red Bandanna Leadership Fellow (0.23% of students)

EMPLOYMENT

Data Analytics Intern, *The Port Authority of New York and New Jersey*, New York, NY May 2026 - August 2026

- Engineered an end-to-end pipeline cross-referencing 2,200+ fleet vehicles against an Oracle SQL database, flagging ~31% as out of compliance with \$100,000+ in outstanding tolls; findings presented to the Deputy Director.
- Built a Python NLP pipeline to language-classify 112,000+ multilingual E-ZPass toll-dispute records, pairing statistical detection (lingua) with quality gates that strip templated and low-signal text to isolate clean English records for downstream LLM analysis.
- Built interactive geospatial tools (Python, Folium, GeoJSON, JavaScript) with cross-filtering, heatmaps, and roadside-image lookups to investigate transponder fraud, surfacing fleet tags operating as far out as Illinois and Florida; adopted by the Revenue Recovery team and reviewed by the agency's Executive Director.

Teaching Assistant, *Boston College Computer Science Department*, Chestnut Hill, MA August 2025 - Present

- Lead weekly discussion sections and daily office hours for 50+ students learning Python fundamentals, translating lecture concepts into worked examples and live debugging.
 - Grade assignments and give targeted feedback on efficiency, readability, and correctness to sharpen students' problem-solving and algorithmic thinking.
 - Partner with the professor to proctor exams, refine course content, and author supplemental coding exercises used across all sections.
-

EXTRACURRICULAR

Student Leader & Big Brother, *BC Bigs*, Chestnut Hill, MA

October 2024 - Present

- Lead biweekly family meetings and coordinate campus-wide events, including "Meet the Bigs Day," building community across Bigs and Littles from multiple partner schools.
 - Mentor a Boston Renaissance Charter School elementary student across 50+ weekly sessions, designing activities that drive academic engagement, teamwork, and personal development.
 - Coordinate with Big Brothers Big Sisters staff and school supervisors to track match progression and resolve scheduling and program issues.
-

PROJECTS

Co-Founder, *W2W-Sports*, W2W-Sports.com

October 2025 - Present

- Constructed a multi-factor watchability scoring algorithm across 10+ sports leagues, weighting variables such as rivalry, marketability, competitiveness, and importance to rank live sporting events in real time.
 - Built a full-stack Flask web application integrating the ESPN API to ingest, process, and score live game data, serving ranked results dynamically to users across multiple sport verticals.
 - Developed an automated email digest system using Resend and Supabase, deploying Vercel Cron jobs to deliver personalized sport rankings to subscribers on a scheduled cadence.
 - Implemented a Pillow-based Open Graph share card generator to produce dynamic image previews per game, increasing content shareability and social media engagement potential.
-

TECHNICAL SKILLS

- Python (Pandas, Numpy), SQL, Power BI, Java, Javascript, HTML & CSS, C, Flask, FastAPI, React, Git, Excel